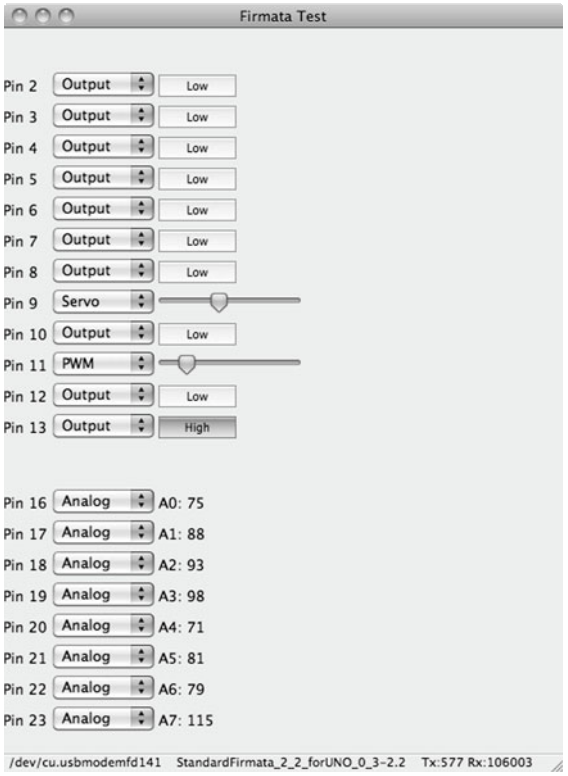


Figure 11-12. *Firmata test program*

With this simple program running, we can configure any of the I/O pins as Input, Output, PWM (for analog output), Servo, and Analog (for analog input). Once configured, we can toggle the state of digital outputs from Low to High, read the state of digital inputs, change the PWM value of PWM pins, move a servo to a different position, and read the values from our Analog inputs. Pretty nifty for a simple little test program, but it also shows some of the versatility in using Firmata with our computer in that it allows us to concentrate on learning a new language without getting bogged down in writing new Arduino code each time we need to do something.

While Firmata host software has been written for a range of software, including Visual Basic, .NET framework, C++/openFrameworks, C#, Perl, Python, Flash, and more, we will look at the use of Firmata on two of the more like-minded programming languages as a way to introduce these packages as candidates for where to go next.

Processing

Processing is the free and open-source programming language that in many ways started things off for the Arduino. Processing was designed by two MIT students, Casey Reas and Ben Fry, as a way to initially make programming visual design and graphics easier for the non-programmer through a text-based language used to generate images. Unlike Arduino, Processing is based off of Java, loosely related to C by